4.6 Formation of Catecholamines

Adrenal glands present in the top of the kidneys are responsible for the secretion of hormones that regulate mineral metabolism (secreted from the adrenal cortex) along with hormones that are secreted in response to stress (secreted by the adrenal medulla), such as epinephrine and nor-epinephrine, that are specialized products of tyrosine (Fig 4.16).

The functions of catecholamines (epinephrine and nor-epinephrine) are:

1. They influence the metabolism of glucose and have an anti-insulin action, i.e. they raise the blood glucose levels.
2. They are responsible for the flight and fight response.
3. These hormones tend to increase the blood pressure by constricting the blood vessels and increasing the force by which heart muscles contract.

The steps involved in the synthesis of catecholamines are:

1. Tyrosine is acted upon by the enzyme tyrosine hydroxylase in the adrenal medulla to form dihydroxy phenyl alanine (DOPA).
2. DOPA is decarboxylated to Dopamine by DOPA decarboxylase.

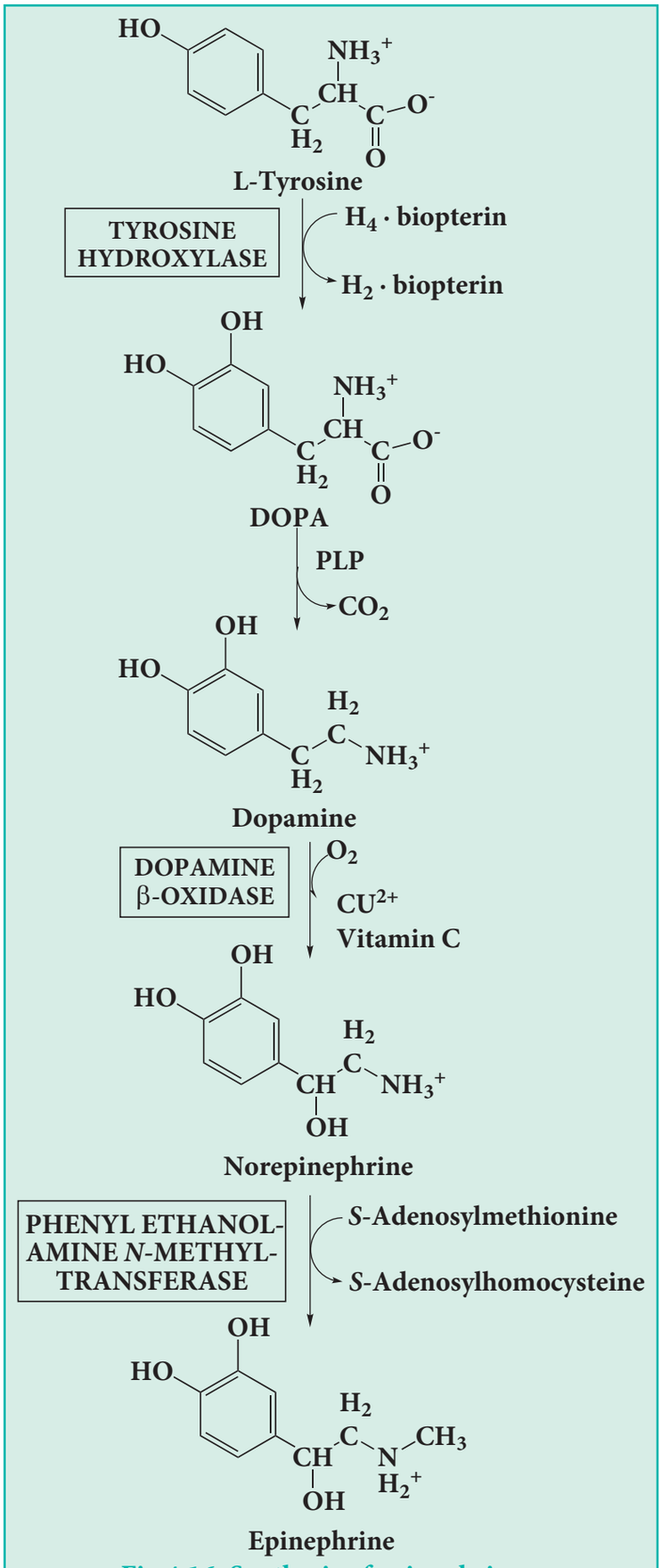

Fig 4.16. Synthesis of epinephrine

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